

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (IT)

Nov 2019

IT471 Foundation of Modern Networking

Date: 13.11.2019, Wednesday

Time: 01:30 pm To 04:30 pm

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1 Answer the following questions:

[07]

a) DNS database contains _____

A - name server records

B - hostname-to-address records

C - hostname aliases

D - all of the mentioned

b) Wildcard domain names start with label _____

A - @

B - *

C - &

D - #

c) The time taken by a packet to travel from client to server and then back to the client is called _____

A - STT

B - RTT

C - PTT

D - JTT

d) Retransmission of packets must be done when

A - Packet is lost

B - Packet is corrupted

C - Packet is needed

D - All of the mentioned

e) Which of the following is connectionless protocol?

A - UDP

B - TCP

C - HTTP

D - ICMP

f) Which of the following statements is NOT true concerning VPNs?

A - Financially rewarding compared to leased lines

B - Allows remote workers to access corporate data

C - Allows LAN-to-LAN connectivity over public networks

D - Is the backbone of the Internet

g) Discarding policy is mainly done by

A - Sender

B - Receiver

C - Router

D - Switch

Q - 2 (a) What is DNS? Explain DNS cache, resolver and name server. How does domain get resolve in IP address? [07]

- (b) What is quality of service? Why it is considered important to measure the network performance? [07]

OR

- Q - 2 (a) What is routing? Explain distance vector routing algorithm. What are the shortcomings of distance vector routing? [07]

- (b) Explain the seven layers of OSI. [07]

- Q - 3 (a) Differentiate between SDN network and conventional network with respect to features, configurations, performance, and innovations [07]

- (b) What is subnetting? How does it work? What are the limitations of existing IP addressing scheme? [07]

OR

- Q - 3 (a) Explain leaky bucket and token bucket algorithm. Also differentiate both. [07]

- (b) Explain control plane and data plane in Software Defined Networking (SDN). How they are advantageous compared to traditional networking? [07]

SECTION – II

- Q - 4 *Answer the following questions:* [07]

- a) **Which of this is not a class of IP address?**

A – Class E

B – Class C

C – Class D

D – Class F

- b) **Which of these is not applicable for IP protocol?**

A - Connectionless

B - Offer reliable service

C - Offer unreliable service

D - None of the mentioned

- c) **What are the queries and responses in ARP?**

A – Query: Broadcast asking about IP

Response: Unicast with MAC address

B – Query: Asking about IP address

Response: Broadcast with MAC address

C – Query: Broadcast asking about MAC address

Response: Unicast with IP address

- d) **Routing within an Autonomous System (AS) is called**

A – Interdomain Routing

B - Intradomain Routing

- e) **Which protocol is responsible for disseminating route inside AS to external destination?**

A - BGP

B - iBGP

C - eBGP

f) How many host IP addresses in a /22 prefix can have?

A - 2^{22}

B - 2^{32}

C - 2^{10}

D - 2^8

g) Which of the following(s) is(are) example of Variable Bit Rate (VBR)?

A - Audio

B - Video

C – Data Transfer

Q - 5 (a) Consider the following characteristics:

[07]

- Variable Delay
- Busy Signals
- Sharing of network resources
- Dedicated resources between sender and receiver.

Categorize the above mention characteristics in packet switching or circuit switching networking with proper reason.

(b) What is NAT? What is end to end argument principle? How does NAT violate end to end argument principle? [07]

OR

Q - 5 (a) Explain spanning tree protocol to eliminate loops from the network with the help of example. [07]

(b) What is Border Gateway Protocol? Differentiate Internal Border Gateway Protocol and External Border Gateway Protocol. [07]

Q – 6 Explain the following in detail (Any two):

[14]

- a) What is lookup algorithm? Categorize the MPLS, Ethernet, ATM, IPv4, IPv6 in exact match and prefix match look up technique. Write the advantage and disadvantage of exact match and prefix match look up technique.
- b) Explain the basic architecture of router along with the task perform by the router. Why each line card having own forwarding table copy considered the important decision?
- c) Explain Internet Protocol (IP) addressing.